



Project : Swasthy Queue

1. Introduction

1.1 Purpose

To provide a digital transparency layer over the physical hospital queue system, allowing patients to monitor their wait time remotely and hospital staff to manage patient flow efficiently.

1.2 Scope

The system involves a patient-facing mobile-responsive web app, a doctor/staff dashboard for calling the "Next" patient, and an admin panel for department management.

2. Overall Description

2.1 Product Perspective

SwasthyaQueue is a Real-Time SaaS platform. It utilizes WebSockets (Socket.io) for instant updates, ensuring that when a doctor clicks "Next," every patient's phone screen updates immediately.

2.2 User Classes

- Patients: Can join a queue via QR code, view live token numbers, and receive notifications.
- Hospital Staff/Reception: Register walk-in patients and assign them to specific doctors.
- Doctors: View their daily list and trigger "Next Patient" or "On Break" status.
- Hospital Admin: Analyze peak hours and department-wise average wait times.

3. System Features (Functional Requirements)

3.1 Digital Token Generation

- Description: Patients can generate a digital token by scanning a QR code placed outside the OPD room or via the hospital portal.
- Logic: The system generates a unique Token_ID based on the department and timestamp.

3.2 Live Queue Tracker (The Core Feature)

- Description: A real-time dashboard showing:
 - Current Token being served.
 - User's own Token number.